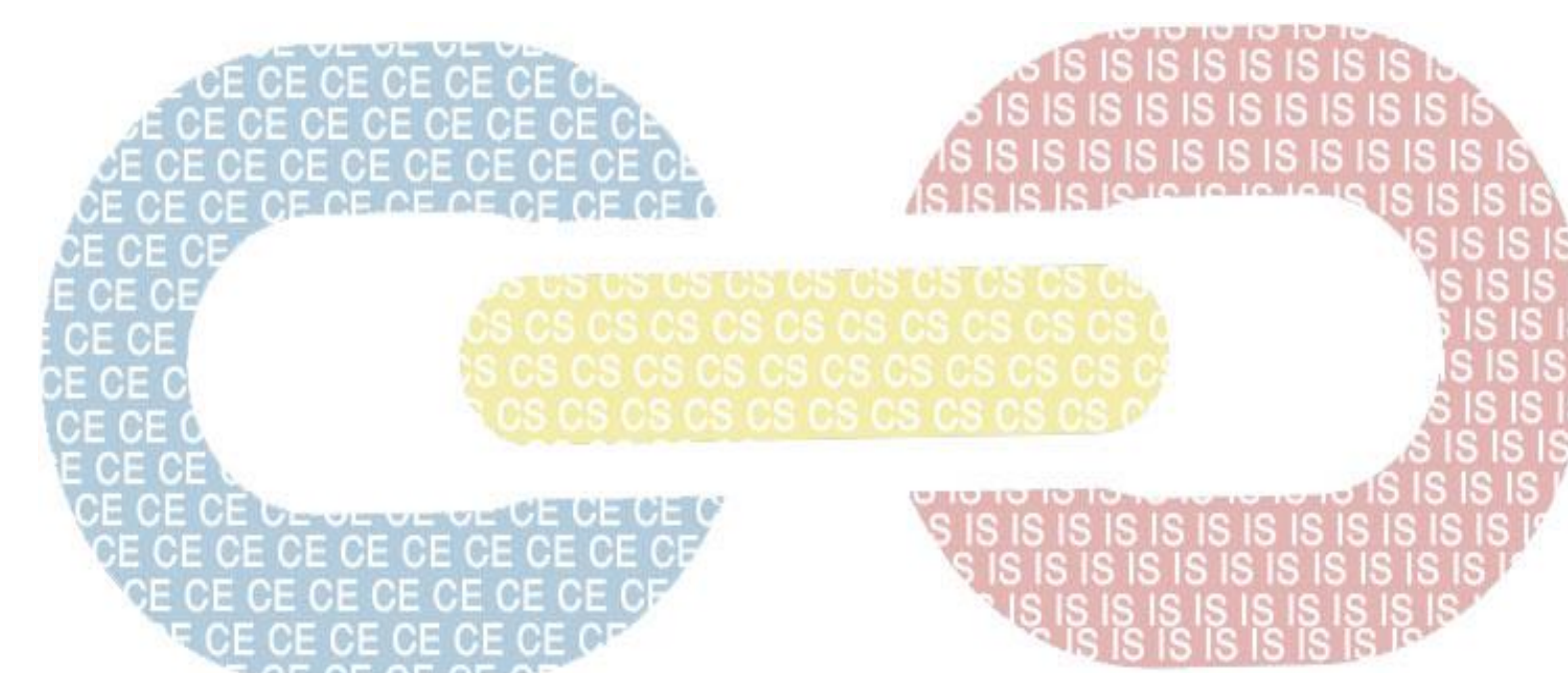




UNIVERSITY OF BAHRAIN

COLLEGE OF INFORMATION TECHNOLOGY

CYBERSECURITY SENIOR PROJECTS



ThaghrAh : Practical Cybersecurity Learning Platform Through Packet Analysis

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ABSTRACT

Beginner cybersecurity learners often struggle to translate theoretical networking knowledge into practical packet-analysis and traffic-investigation skills. Many educational platforms provide limited exposure to realistic network traffic and hands-on cybersecurity tools, which can reduce learner confidence and practical understanding. This project introduces ThaghrAh, a web-based cybersecurity learning platform designed to support practical packet analysis, protocol investigation, and Wireshark-based training through downloadable PCAP challenges and AI-generated investigation scenarios.

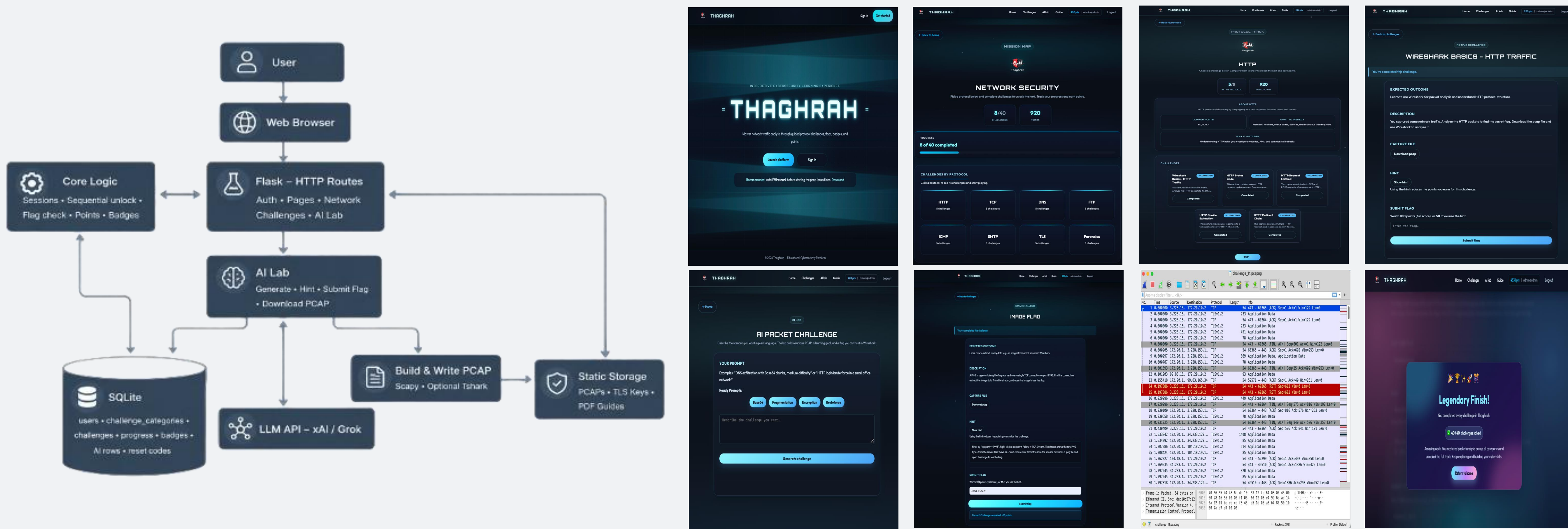
The platform was implemented using Flask, SQLite, Scapy, and configurable AI-provider integration. ThaghrAh provides a structured learning environment consisting of 40 packet-analysis challenges distributed across multiple protocol categories, alongside an AI Lab capable of generating dynamic investigation scenarios from natural-language prompts. The platform incorporates guided challenge progression, automated flag validation, gamified scoring and badge mechanisms, protocol learning resources, and practical PCAP-based investigation workflows.

Evaluation findings indicate that combining structured progression, practical investigation activities, and gamified learning features can improve learner engagement and strengthen practical cybersecurity education. The project also demonstrates that AI-generated investigation scenarios can increase training diversity while maintaining realistic packet-analysis workflows within a controlled educational environment.

OBJECTIVES

- Help beginner cybersecurity learners develop practical packet-analysis skills.
- Bridge the gap between theoretical networking knowledge and hands-on investigation.
- Provide realistic PCAP-based cybersecurity challenges through protocol-focused practical exercises.
- Encourage independent analysis, critical thinking, and self-driven problem-solving by applying reduced scoring when hints are used.
- Support learner progression through structured challenge sequencing, progress tracking, and gamified reward mechanisms.
- Integrate AI-generated investigation scenarios to increase challenge diversity and training realism.

SYSTEM ARCHITECTURE



RESULTS



CONCLUSION & FUTURE WORK

- Conclusion:**
- Evaluation confirmed successful operation within the intended educational environment.
 - The platform supported challenge progression, flag validation, PCAP workflows, security protections, and AI-generated challenges.
 - ThaghrAh effectively delivered practical packet-analysis learning through guided cybersecurity investigation.
- Future Work:**
- Expand challenge coverage to broader cybersecurity domains.
 - Improve the quality and consistency of AI-generated scenarios.
 - Introduce team-based competition features such as leaderboards and collaborative challenges.
 - Add real-time packet capture and live traffic analysis.